

## WSTLUR 2026

### Can We Trust OpenStreetMap to Measure Cycling-Infrastructure Change?

#### Slide 1 — Title

Nǐ hǎo (ni jau) - Good afternoon everyone.

My name is Eugeni Vidal-Tortosa, and I'm a researcher at GEMOTT, the Research Group on Mobility, Transport, and Territory at the Universitat Autònoma de Barcelona.

I am here today to share some work we've been doing together with Oriol Marquet and Víctor González-Parra as part of the ATRAPA project, which is an ERC-funded project on the acceptability of urban transformations.

The question driving this work is simple: can we trust OpenStreetMap to measure cycling infrastructure change?

#### Slide 2 — Why does this matter?

Let me start with why this question is important.

Cities around the world are rapidly expanding cycling infrastructure to improve public health, reduce environmental impacts, and make cities more liveable. These images, for example, show new cycling infrastructure in Barcelona, but similar changes are happening in Paris, Bogotá, here in Beijing, and many other cities.

But what impact is this new cycling infrastructure having? Does it lead to more cycling? Does it reduce car use? Or (as we ask in the ATRAPA project) does it affect electoral support? To answer these kinds of questions, we need longitudinal data. In other words, data not only on the infrastructure that exists today, but also on how it has changed over time. However, longitudinal data are often incomplete, inconsistent across cities, or unavailable.

This is where OpenStreetMap, or OSM, comes in, because OSM allows us to download global, historical snapshots from sources such as Geofabrik and compare cycling networks across different points in time.

But because OSM is built by volunteers, this raises the crucial question: can we trust it to detect these changes over time?

### **Slide 3 — Research aim**

So, our research aim is to quantify the accuracy of OSM for detecting cycling-infrastructure change.

To achieve this, we developed a reproducible open-source validation framework.

### **Slide 4 — Study setting and data sources**

We applied this framework to Barcelona between 2015 and 2023, an eight-year period during which the city expanded its cycling network substantially.

To implement it, we combined three main data sources: historical OSM cycling-infrastructure snapshots (which you can see on the maps), census-tract boundaries to support the sampling design, and historical Google Street View, or GSV, imagery used as a proxy for ground truth.

### **Slide 5 — Analytical workflow**

The framework follows five steps.

First, we detect cycling-infrastructure changes in OSM, including both additions and removals. Second, we stratify the census-tract sample to ensure representation across different urban contexts. Third, we generate network validation points within the selected census tracts. Fourth, we inspect and code historical GSV imagery at those points. And finally, we calculate the validation metrics.

Let's now go through each of these steps one by one.

### **Slide 6 — Detected cycling-infrastructure changes**

Starting with step one. We applied a geometric differencing approach to identify cycling-infrastructure changes between the two OSM snapshots, from 2015 and 2023.

The results can be seen on this map.

Overall, we identified around 127 kilometres of additions and 25 kilometres of removals, resulting in an 80% increase in the cycling network.

In total, this gave us 946 candidate change segments, which formed the basis for the validation exercise.

### **Slide 7 — Stratified validation design**

In step two, to ensure representation across different urban contexts, we stratified Barcelona by population density and centrality, creating nine strata.

And as shown on the bi-variate map, we then randomly selected six census tracts from each stratum, giving us a total of 54 census tracts.

### **Slide 8 — Validation points**

Step three. Within each of those 54 census tracts, we sampled up to five validation points, attempting to include two OSM-detected additions, two OSM-detected removals, and one control location where no cycling infrastructure was present in either year.

We then linked each point to a before and an after GSV image, corresponding to around 2015 and 2023.

### **Slide 9 — Inspecting and coding historical GSV imagery**

Step four is where the validation happens.

For each validation point, we compare the before and after imagery and check whether the change identified by OSM actually occurred.

In the top row, we can see an example of an OSM-detected addition. In the before image there is no cycling infrastructure, whereas in the after image a new segregated cycle track has been installed, confirming the addition.

In the bottom row, we can see an OSM-detected removal. In the before image, a red cycle lane is visible, but in the after image the street has been redesigned and pedestrianised, confirming the removal.

These are examples where OSM detected the change correctly. But that is not always the case. That's why validation is necessary.

### **Slide 10 — Validation metrics**

In the final step, we summarise the results using two standard validation metrics: precision and recall.

Precision tells us what proportion of the changes detected by OSM were actually real.

Recall tells us what proportion of real changes were captured by OSM.

Together, these metrics allow us to assess how well OSM captures infrastructure change.

## Slide 11 — Validation performance

Ok, so what were our results?

For additions, which represent most of the detected changes, OSM performs very well. Precision was 0.82 (zero point eight two), meaning 82% of the recorded additions were confirmed by GSV. Recall was 0.96, meaning that OSM captured 96% of the real additions identified in the imagery.

For removals, the picture was much less clear. We were only able to validate five cases, resulting in highly uncertain estimates and wide confidence intervals. Precision was 0.20 and recall 1.00, but these values should be interpreted very cautiously.

Looking at the pooled results, precision was 0.74 and recall was 0.97. However, these results are dominated by additions, which represent most of the detected changes.

## Slide 12 — Limitations

A few limitations to keep in mind.

First, our validation relied on historical GSV imagery, which is only a proxy for ground truth. Coverage is uneven across space and time, and as a result, some locations could not be validated.

Second, because historical GSV imagery is not accessible through the GSV API, validation had to be carried out manually, making the process time-consuming and difficult to scale.

Third, the removal estimates remained uncertain due to the small number of cases. We tried to increase the sample, but removals were rare, concentrated in a few areas, and often lacked suitable historical GSV imagery.

And fourth, we know that Barcelona has a strong OSM community and is likely better mapped than many other cities. So we may not see the same results elsewhere, and more work is needed to understand how well these findings transfer to other settings.

## Slide 13 — Conclusions

So, taking all of this together, and coming back to our original research question, can we trust OSM to measure cycling-infrastructure change?

The answer depends on the type of change we are interested in.

For additions, the evidence is quite strong. OSM appears to capture cycling-network expansion reasonably well, making it a useful source for tracking network growth, particularly where consistent historical records are unavailable.

For removals, however, the picture is much less clear. Removals were relatively rare, making them difficult to validate, and most apparent removals reflected mapping edits rather than real infrastructure loss. These changes should therefore be interpreted cautiously and verified independently whenever possible.

More broadly, OSM accuracy should not be assumed. Performance depends on the type of change being studied, the local mapping context, and the available validation data.

Our main contribution is a reproducible framework that allows researchers to measure, evaluate, and calibrate OSM-derived estimates of cycling-infrastructure change before using them in longitudinal analyses. And, in fact, as part of the ATRAPA project, we are currently using this framework to evaluate OSM-derived infrastructure change in other European cities.

## Slide 14 — Thank you

Xièxie (siè sie)- Thank you very much.

I'd be happy to take any questions, and also to share the code if anyone is interested.